Tutorial 2

Learning, cascades, and games

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How this tutorial works

- Four blocks. Each one opens with a compact recap of the theory and a pointer to the lecture where it is developed, and then moves to exercises on that same material.
- The exercises use the same templates as the online quizzes on Virtuale and as the mid-term. Same mechanics, different numbers.
- Try each one first. We solve it together afterwards.
- Questions on the *content* of the course go to Prof. Bigoni. Questions on the organisation, the quizzes and the exam come to me: `marco.rosso4@unibo.it`

Coverage: Lectures 5, 6, 7, 8.

Plan

1. Learning and information cascades
2. Simultaneous games: prisoner's dilemma and coordination
3. Expectations, guessing games, mixed strategies
4. Multi-stage games

Sample *oral* questions at the end.

PART 1

Learning and information cascades

Belief learning and reinforcement learning

Lecture 5 – "Belief learning", "Reinforcement learning", "Belief vs. Reinforcement learning"

$$\text{belief learning} \qquad \Pr(L) = \frac{\alpha + N_L}{\alpha + \alpha + N} \qquad N = N_L + N_R$$

$$\text{reinforcement learning} \qquad \Pr(L) = \frac{\alpha + e_L}{2\alpha + e_L + e_R}$$

- α determines how quickly probabilities respond: a large α keeps them close to $1/2$.
- Belief learning: N_L, N_R are the **observations** of each event. The left-hand side is a belief.
- Reinforcement learning: e_L, e_R are the total **earnings** from predicting each event. The left-hand side is a choice probability.

Exercise 1.1

Learning

(a) In a binary prediction task, a subject observes the sequence of events

$$L,\ L,\ L,\ L,\ R.$$

Using the belief-learning model with $\alpha = 2$, compute $\Pr(L)$ after these five observations.

(b) Another subject earns €1 for each correct prediction, nothing otherwise. After 10 trials she has predicted L 7 times (5 of them correct) and R 3 times (1 correct). Under reinforcement learning with $\alpha = 2$, what is the probability that she predicts L in trial 11?

Information cascades

Lecture 5 – "The basic situation" to "Incorrect cascades"

Two cups, equally likely. Each person sees **one private draw** and **all previous decisions**, not previous draws. Paid only for guessing the cup.

- 1st person: follows own draw.
  $\Pr(A \mid a) = 2/3$
- 2nd person: in a conflict, the two draws cancel out $\Rightarrow$ goes with own draw
- From the 3rd on: two matching decisions outweigh one's own draw $\Rightarrow$ **cascade**

prior	own	$\Pr(A)$	decision	
A, A	*a*	0.89	A	
A, A	*b*	0.67	A	*(cascade)*
A, B	*a*	0.67	A	
A, B	*b*	0.33	B	

Incorrect cascade from the first two draws: $\left(\frac{1}{3}\right)\left(\frac{1}{3}\right) = \frac{1}{9}$. In a cascade, decisions convey **no information** about private draws $\Rightarrow$ cascades may be fragile.

Exercise 1.2

Information cascades

Cup A contains **3 Amber and 1 Blue**; Cup B contains **1 Amber and 3 Blue**. Each cup is equally likely. People decide in sequence, each seeing one private draw (with replacement) and all previous decisions.

(a) The first person draws amber. What is her probability of Cup A?
(b) Complete the table for the third person, assuming that the first two followed their own draws:

prior decisions	own draw	Pr(Cup A)	decision
A, A	*a*	_____	____
A, A	*b*	_____	____
A, B	*a*	_____	____
A, B	*b*	_____	____

(c) Which row starts a cascade?
(d) What is the probability that the first two people both receive a misleading draw, and so start an incorrect cascade?

PART 2

Simultaneous games

Nash equilibrium and the prisoner's dilemma

Lecture 6 – "A game", "Nash equilibrium", "The prisoner's dilemma", "Backward induction"

Nash equilibrium: a set of strategies, one for each player, such that nobody would wish to deviate, given the strategies used by the others.

Method: underline each player's best response to each strategy of the other. A cell with both payoffs underlined is a NE.

	Y	Z
Y	$\underline{3}, \underline{3}$	$\underline{10}, 0$
Z	$0, \underline{10}$	$8, 8$

- Whatever the other does, Y is better: (Y,Y) is the only NE, although (Z,Z) is better for both
- Finite, known number of rounds: backward induction $\Rightarrow$ no cooperation, even in the first round

Lab: random matching, 43% cooperation early $\rightarrow$ about 20% in rounds 15–20. Partner matching: more cooperation, which drops as the final round approaches (*end-game effect*).

Exercise 2.1

A game

Consider a game with the following payoff matrix.

	Left	Right
Top	35, 35	10, 45
Bottom	45, 10	20, 20

(a) Identify the pure-strategy equilibrium outcomes of this game: a. (35, 35) b. (10, 45) c. (45, 10) d. (20, 20)

(b) What type of game is this?

a. prisoners' dilemma b. battle of the sexes c. ultimatum game d. matching pennies game e. traveler's dilemma

Exercise 2.2

Repetition

Consider the game of Exercise 2.1.

(a) What is the Nash equilibrium of the game, when it is repeated for 10 periods with fixed pairs? Briefly illustrate the reasoning behind your answer.

(b) Consider an experiment in which the game is played for 10 periods, with fixed pairs. What is the behaviour that normally emerges in this type of situation, in the laboratory?

From production to coordination

Lecture 6 – “Alternative framing: co-workers”, “Coordination problems”, “Equilibrium selection”

Two workers, effort 0 or 1. The benefit per person depends on total effort; each unit of effort costs the worker who supplies it.

$$\text{payoff} = \text{benefit per person}(\text{total effort}) - \text{cost} \times \text{own effort}$$

total effort	0	1	2	cost	game
no synergy	3	10	18	10	prisoner’s dilemma: (Y,Y)
synergy	3	10	30	10	coordination: (Y,Y) and (Z,Z)
high cost	3	10	30	19	coordination, (Z,Z) very risky

Coordinating on the best equilibrium is not guaranteed: with a high cost of effort, behaviour is likely to converge to the equilibrium that is worst for all.

Exercise 2.3

Co-workers

Two co-workers each choose effort L (0) or H (1). The benefit per person depends on total effort:

total effort	0	1	2
benefit per person	€4	€12	€36

Each unit of effort costs €c to the worker who supplies it.

(a) Write the payoff matrix for $c = 12$ and find all pure-strategy Nash equilibria.

(b) Same, for $c = 20$.

(c) Now suppose there is no synergy: the benefit with total effort 2 is €20, and $c = 12$. What type of game is it?

PART 3

Expectations, guessing games, mixed strategies

Expected payoff difference and basins of attraction

p = probability that the other player chooses High.

$$\Delta = \mathbb{E}[\pi_H] - \mathbb{E}[\pi_L] \qquad \text{threshold: } \Delta = 0$$

game	Δ	threshold
(3,3), (10,0), (0,10), (20,20)	$13p - 3$	$p = 3/13 \approx 0.23$
(3,3), (10,−7), (−7,10), (13,13)	$13p - 10$	$p = 10/13 \approx 0.77$

- A risk-neutral player prefers High when p is above the threshold.
- **Basin of attraction** of an equilibrium: the beliefs for which learning pushes towards it. A higher cost of effort raises the threshold and shrinks the basin of the high-effort equilibrium.

Exercise 3.1

Thresholds

Take the two coordination games of Exercise 2.3, with $c = 12$ and $c = 20$. Let p be the probability that the other worker chooses H.

(a) For each game, write $\mathbb{E}[\pi_H]$, $\mathbb{E}[\pi_L]$ and Δ.

(b) How large should p be, for a risk-neutral worker to prefer H?

(c) What happens to the basin of attraction of (H,H) when the cost rises?

Guessing games

Lecture 7 – “The game”, “The Nash equilibrium”, “Experimental results”, “Introspection”

N players pick a number from 0 to 100. The winner is the one closest to $1/2$ of the average.

- The best response to any common decision above 0 is to go below it $\Rightarrow$ **the only Nash equilibrium is 0**.
- **Level-k**: level 0 does not think (randomises); level 1 best responds to level 0; level k best responds to level $k-1$. Most people are level 1 to 3.
- Lecture: three levels of iterated reasoning give $50 \to 25 \to 12.5 \to 6.25$.
- With repetition, people learn and choices converge towards the Nash equilibrium.

Exercise 3.2

Beauty contest

Five players each choose a number between 0 and 100. The winner is the player whose number is closest to **2/3** of the average of all numbers.

(a) What is the Nash equilibrium?

(b) Level-0 players choose at random, so that their average is 50. What do level-1, level-2 and level-3 players choose?

(c) The five choices are 40, 30, 25, 18 and 12. Who wins?

Exercise 3.3

A two-player game

The Row player can choose between “up” and “down”; the Column player can choose between “left” and “right”.

- If Row chooses up and Column chooses left: Row earns 8, Column earns 2.
- If Row chooses down and Column chooses right: Row earns 4, Column earns 6.
- In all other cases both players earn 0.

(a) Write the payoff table for this game.

(b) Select which category this game belongs to: a. prisoner’s dilemma b. matching pennies game c. battle of the sexes d. trust game

(c) Select all the Nash equilibria in pure strategies: a. DOWN–RIGHT b. DOWN–LEFT c. UP–LEFT d. UP–RIGHT

Mixed strategies

A player randomises only if she is **indifferent** between the actions she mixes over.

p = Pr(Column plays Right), q = Pr(Row plays Up).

- Row’s expected payoffs depend on p; Column’s depend on q.
- Mixed equilibrium: p makes Row indifferent, $\mathbb{E}[\pi_{\text{up}}] = \mathbb{E}[\pi_{\text{down}}]$; q makes Column indifferent, $\mathbb{E}[\pi_{\text{left}}] = \mathbb{E}[\pi_{\text{right}}]$.

Best-response functions: Row plays Up ($q = 1$) on one side of the indifference value of p and Down ($q = 0$) on the other, and symmetrically for Column. Their intersections are *all* the Nash equilibria, pure and mixed.

Exercise 3.4

Expected profits and mixed strategies

Same game as Exercise 3.3. Let p denote the probability that the Column player chooses "right", and q the probability that the Row player chooses "up".

(a) Complete the following equations for the expected profits of each player's possible choices:

$$\mathbb{E}[\pi_{\text{up}}] = \square\,(1-\square) + \square \cdot \square \qquad \mathbb{E}[\pi_{\text{left}}] = \square \cdot \square + \square\,(1-\square)$$
$$\mathbb{E}[\pi_{\text{down}}] = \square\,(1-\square) + \square \cdot \square \qquad \mathbb{E}[\pi_{\text{right}}] = \square \cdot \square + \square\,(1-\square)$$

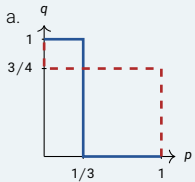
(b) The equilibrium in mixed strategies prescribes that the probability that Column plays right is $p = ?$, and the probability that Row plays up is $q = ?$

Exercise 3.5

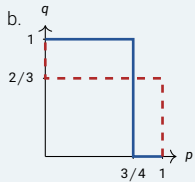
Best-response functions

Same game. Select the graph that gives the correct representation of the best-response functions (solid: Row's best response, dashed: Column's best response).

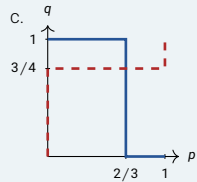
a.



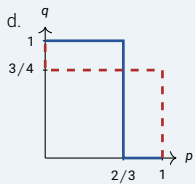
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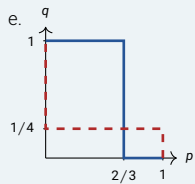
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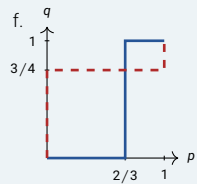
d.



e.



f.



PART 4

Multi-stage games

Strategies, Nash equilibrium, subgame perfection

Lecture 8 – “Strategies”, “Nash equilibria”, “Extensive form”, “Non-credible threats”, “Subgame-perfection”, “Backward induction”

- **Strategy**: a complete plan of action, covering every contingency, including decision points that are never reached.
- Lecture: with 2 proposals, the responder has 4 strategies (AA, AR, RA, RR), one reaction to each proposal.
- A **Nash equilibrium** may rest on a *non-credible threat*: (Equal, RA) is a NE, but rejecting the unequal split would reduce the responder’s payoff from $1 to $0.
- **Subgame perfect NE**: rationality in all parts of the game (subgames). Found by **backward induction**, starting from the final decision.
- Lecture: the only SPNE is (Unequal, AA).

Exercise 4.1

Bargaining – extensive form

Consider a game in which there is €6 to be divided, and the first mover is only permitted to make one of three proposals:

- LOW: €5 for the first mover and €1 for the second mover,
- EQUAL: €3 for each, or
- HIGH: €1 for the first mover and €5 for the second mover.

The second mover is shown the proposal and can either accept, in which case it is implemented, or reject and cause each to earn €0.

(a) Show this game in extensive form. How many subgames are there?

(b) How many strategies does each player have?

Exercise 4.2

Bargaining – a Nash equilibrium

Show that proposal HIGH in the game of Exercise 4.1 is part of a Nash equilibrium.

To finish specifying the decisions for this equilibrium, you have to say what the second mover's response is to each of the three proposals.

Exercise 4.3

Sequential rationality

(a) Does behaviour in the equilibrium discussed in Exercise 4.2 satisfy sequential rationality?
a. No b. Yes

(b) Find the subgame perfect Nash equilibrium of the game by backward induction.

Centipede game and noisy choice

Lecture 8 – “A treasure”, “A contradiction”, “The centipede game”, “The logit model”, “QRE and the non-credible threat”

Centipede (4 legs): backward induction $\Rightarrow$ Red stops at the first node, $(40, 10)$. McKelvey and Palfrey: only **8%** of Red players stop there.

Treasure vs contradiction: same SPNE (R, N), but when punishing (P) costs the second mover only 2, more than half of the first movers choose S.

Logit, with noise parameter λ:

$$\Pr(D_1) = \frac{\exp(\pi_1 \lambda)}{\exp(\pi_1 \lambda) + \exp(\pi_2 \lambda)}$$

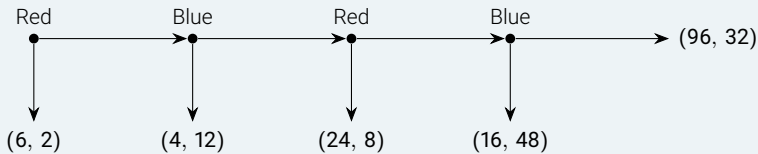
Lecture, contradiction game, $\lambda = 0.1$: $\Pr(P) = \dfrac{\exp(0.1 \times 48)}{\exp(0.1 \times 48) + \exp(0.1 \times 50)} = 0.45$.

QRE: better strategies are played more often, but not always; and expectations in equilibrium are correct.

Exercise 4.4

Centipede

At each node the player moving can *stop* (down), which ends the game, or *pass* (right). Payoffs are (Red, Blue).

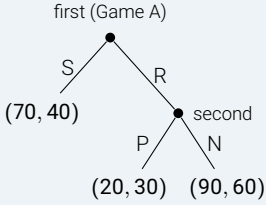


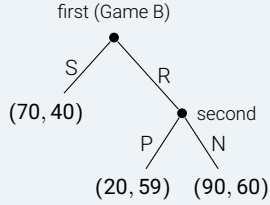
(a) Find the subgame perfect equilibrium outcome by backward induction.

(b) What do experiments find in this type of game, and why?

Exercise 4.5

Noisy best responses





(a) Find the subgame perfect Nash equilibrium of each game. Is there a Nash equilibrium that is not subgame perfect?

(b) The second mover chooses according to the logit model with $\lambda = 0.1$. Compute the probability of P in each game.

PART 5

Oral questions

Sample oral questions I

Information cascades

1. Can you explain what an information cascade is?
2. Describe the experiment in which information cascades were studied.
3. Are information cascades compatible with the standard assumptions on perfect rationality, or do they only emerge if people are irrational?
4. Can a rational cascade induce the widespread adoption of a systematically incorrect choice?

Learning and simultaneous games

1. What is the difference between belief learning and reinforcement learning?
2. What happens to cooperation in a prisoner's dilemma repeated a known number of times, in theory and in the laboratory?
3. Describe the beauty contest game. What is its Nash equilibrium, and what do experiments show?

Sample oral questions II

Multi-stage games

1. What is a non-credible threat? Give an example.
2. What is the difference between a Nash equilibrium and a subgame perfect Nash equilibrium?
3. Describe the centipede game. What does backward induction predict, and what is observed?
4. How can a model of noisy best responses (logit, QRE) explain behaviour in games where the cost of punishment is very small?